13. Regression Discontinuity Design

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Regression Discontinuity Design* (Prof. Sanghack Lee, GSDS SNU; adapted from Ed Rubin, Matthew Blackwell, Fan Li and Kosuke Imai)

The central question is how a known cutoff rule for treatment assignment identifies a causal effect at the cutoff, how that effect is estimated from data near the boundary, and what the bandwidth choice does to bias, variance and inference.

These notes consolidate the slides, prove the identification result, derive the local-linear estimators, explain the MSE- and coverage-optimal bandwidths, treat the fuzzy design as a local IV, and summarize the Chicago remedial-education case study. Physical PDF page numbers are listed in Section F.

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1 Where RDD fits

Purpose. Place RDD among empirical designs.

Types of empirical studies, from more to less researcher control:

Experiment (RCT)	Natural experiment	Quasi-experiment	Observational study
researcher randomizes treatment	as-if random assignment by nature/policy (lottery, IV)	known non-random rule exploited (RD, DiD, SCM)	confounders must be assumed away (matching, IPW)

Method	Design type	Key source of variation
Back-door / matching / IPW	observational	adjusted confounders
IV	natural / quasi-experiment	exogenous instrument
RDD	quasi-experiment	cutoff rule
DiD	quasi-experiment	temporal variation across groups
SCM	quasi-experiment	weighted donor pool

IV straddles the boundary: the instrument may come from a natural experiment (lottery) or a quasi-experimental rule (policy threshold). From here on the focus is on quasi-experimental designs that exploit known institutional rules. So far: back-door adjustment (block non-causal paths) and IV (external instrument under unobserved confounding). Today: treatment determined by whether a pre-treatment variable crosses a known threshold.

Notation. $Z_i \in \mathbb{R}$: forcing (running) variable (not an instrument as in Lecture 11-2); c: cutoff; $X_i = \mathbb{1}\{Z_i \geq c\}$: treatment in the sharp design; $Y_i(1), Y_i(0)$: potential outcomes; τ_{SRD}: sharp-RD effect at the cutoff; $W_i = \mathbb{1}\{Z_i \geq c\}$: the instrument in the fuzzy design (playing the role of Z_i in the IV lecture).

2 Concept and identification

Purpose. Define the sharp RD effect and show that continuity identifies it.

2.1 Examples

Incumbency advantage (Lee 2008; Lee et al. 2004). Does winning an election raise the chance of winning the next? Running variable: Democratic vote-share margin at election t; cutoff: 50%; outcome: Democratic vote share at $t + 1$ (or the ADA score of the representative). Candidates who barely win and barely lose are comparable in observables and unobservables; near the cutoff the election outcome is as good as random. Barely winning causes a 6–8 percentage-point increase in next-election vote share: a substantial incumbency advantage. Often regarded as the gold-standard RDD application.

Scholarships and test scores (Thistlethwaite and Campbell 1960, the first RDD): eligibility depends solely on whether an exam score exceeds a threshold; students just above and below are arguably comparable in ability and attitudes. **Financial aid and family income**: eligibility depends on family income below a threshold. In both, the cutoff is known and pre-determined and units near it are plausibly comparable.

2.2 The design

In an RDD, treatment status changes discontinuously in a pre-treatment forcing variable, while unobserved confounders vary smoothly around the cutoff. Knowledge of the assignment mechanism identifies a *local* causal effect at the cutoff, by comparing units with similar forcing values but different treatment. The discontinuity is often created by an artificial, pre-fixed threshold.

Sharp RD. Z_i continuous (discrete running variables need extra care; Cattaneo et al. 2020) and $X_i = \mathbb{1}\{Z_i \geq c\}$, a deterministic function of the forcing variable (scores above 1500 get the scholarship). Positivity is violated by construction:

$$P[X_i = 1 \mid Z_i = c - \varepsilon] = 0, \qquad P[X_i = 1 \mid Z_i = c + \varepsilon] = 1,$$

so back-door adjustment and IPW cannot deliver the ATE/ATT; continuity-based extrapolation at the cutoff is used instead.

Fuzzy RD. Crossing the threshold changes the *probability* of treatment discontinuously, but not from 0 to 1 (Section 5).

2.3 Estimand and identification

$$\tau_{\text{SRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = c] = \mathbb{E}[Y_i(1) \mid Z_i = c] - \mathbb{E}[Y_i(0) \mid Z_i = c].$$

Since Z_i is continuous, no unit has $Z_i = c$ exactly; identification relies on local extrapolation, which needs an assumption.

Continuity assumption. The CEFs $\mu_1(z) = \mathbb{E}[Y_i(1) \mid Z_i = z]$ and $\mu_0(z) = \mathbb{E}[Y_i(0) \mid Z_i = z]$ are continuous in z. Then

$$\begin{aligned}
\mathbb{E}[Y_i(0) \mid Z_i = c] &= \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid Z_i = z] && \text{(continuity)} \\
&= \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid X_i = 0, Z_i = z] && \text{(SRD: } z < c \Rightarrow X_i = 0\text{)} \\
&= \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z] && \text{(consistency),}
\end{aligned}$$

and likewise $\mathbb{E}[Y_i(1) \mid Z_i = c] = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$. Hence

$$\boxed{\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z].}$$

The estimation problem is to estimate two regression functions at a boundary point: very hard without parametric assumptions; nonparametric regression is consistent but converges slowly.

What can go wrong? Sorting around the threshold violates smoothness: students retake an exam to pass the aid threshold; richer students retake more often, so those just above differ systematically from those just below.

3 Validity and diagnostics

Purpose. Check the design's assumptions graphically and with falsification tests.

Most RD analyses show some of three figure types:

1. **Outcomes by running variable:** is there a jump at the cutoff? "An RDD is hard to sell if the effect is not visually obvious in the outcome plot" (Imbens and Lemieux 2008).

2. **Covariates by running variable:** covariates should be smooth across c; a discontinuity in covariate means suggests discontinuities in potential-outcome means, and the RD may be invalid.
3. **Density of the running variable:** is there evidence of sorting? Estimate the density separately on each side; a discontinuity suggests sorting.

Manipulation and the McCrary test. RDD assumes units cannot precisely manipulate Z_i to land on a preferred side. If they can, smoothness of potential outcomes is likely violated. McCrary (2008) tests for a discontinuity of the density $f_Z(z)$ at c; a significant jump suggests manipulation, in which case units just above and below are no longer comparable.

Bunching. With administrative data, individuals or firms are often seen to locate at policy thresholds: firms reporting earnings just below a tax or regulation threshold, workers choosing hours just below the full-time cutoff (World Bank blog "Ready, set, bunch"). Example: the distribution of reported wealth in Colombia before (2009) and after (2010) a wealth-tax reform shows bunching just below the new threshold (Londoño-Vélez).

Falsification (placebo) tests. If the RDD is valid, the effect appears only at the true cutoff.

- *Placebo cutoffs:* re-estimate at fake cutoffs away from c; with smooth CEFs no jump should appear. In a simulation with an effect only at $c = 0$, estimates at placebo cutoffs between -2 and 2 are near zero with CIs covering zero, while the true cutoff shows a clear effect.
- *Placebo outcomes:* use pre-treatment covariates as outcomes; treatment cannot affect them, so a jump signals sorting.

4 Estimation

Purpose. Estimate boundary limits with local polynomial regression and choose the bandwidth.

4.1 General strategy

The goal is limits such as $\lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$. Two differences from standard nonparametric regression: the target is a single point, and it is a *boundary* point where data exist on one side only, so boundary bias shrinks slowly. The naive difference in means uses data too far from the boundary.

Upper and lower limits: $\mu_+(c) = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$, $\mu_-(c) = \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]$, and $\tau_{\text{SRD}} = \mu_+(c) - \mu_-(c)$.

Kernel (local average) regression. With a uniform kernel and bandwidth h,

$$\hat{\mu}_-(c) = \frac{\sum_{i=1}^n Y_i \mathbb{1}\{c - h \le Z_i < c\}}{\sum_{i=1}^n \mathbb{1}\{c - h \le Z_i < c\}},$$

the mean among units at most h below the threshold (similarly above). Equivalently, regress on units within h of c:

$$(\hat{\alpha}, \hat{\tau}_{\text{SRD}}) = \arg\min_{\alpha, \tau} \sum_{i : Z_i \in [c-h, c+h]} (Y_i - \alpha - \tau X_i)^2,$$

where, with binary X_i, the OLS coefficient is the difference in means. This assumes locally constant potential outcomes on each side. h governs the bias–variance trade-off: large h, high bias and low variance; small h, low bias and high variance. Bias shrinks slowly in h, so finite-sample bias may be large. Kernels: uniform $K(u) = \frac{1}{2}\mathbb{1}\{|u| < 1\}$, triangular $K(u) = (1 - |u|)\mathbb{1}\{|u| < 1\}$.

4.2 Local linear regression

With a wider bandwidth the CEF may not be flat, so allow a slope. On $Z_i \in [c-h, c)$,

$$(\hat{\alpha}_-, \hat{\beta}_-) = \arg\min_{\alpha,\beta} \sum_{i:Z_i\in[c-h,c)} \big(Y_i - \alpha - \beta(Z_i - c)\big)^2,$$

and the same on $[c, c+h]$. Then

$$\hat{\tau}_{\text{SRD}} = \hat{\mu}_+(c) - \hat{\mu}_-(c) = \hat{\alpha}_+ + \hat{\beta}_+(c-c) - \hat{\alpha}_- - \hat{\beta}_-(c-c) = \hat{\alpha}_+ - \hat{\alpha}_-.$$

Single regression. Equivalently,

$$\arg\min_{(\alpha,\beta,\tau,\gamma)} \sum_{i:Z_i\in[c-h,c+h]} \big\{Y_i - \alpha - \beta(Z_i - c) - \tau X_i - \gamma(Z_i - c)X_i\big\}^2,$$

with $\hat{\tau}_{\text{SRD}} = \hat{\tau}$. For $X_i = 0$ the fit is $\alpha + \beta(Z_i - c)$; for $X_i = 1$ it is $(\alpha+\tau) + (\beta+\gamma)(Z_i - c)$: one regression on top of the other.

Kernel weighting. Weight points near c more heavily:

$$\arg\min_{(\alpha,\beta,\tau,\gamma)} \sum_{i=1}^{n} K\left(\frac{Z_i - c}{h}\right) \big\{Y_i - \alpha - \beta(Z_i - c) - \tau X_i - \gamma(Z_i - c)X_i\big\}^2,$$

typically with the triangular kernel.

4.3 Bias–variance trade-off and the MSE-optimal bandwidth

Let $\mu(z) = \mathbb{E}[Y \mid Z = z]$ with one-sided second derivatives $\mu_\pm^{(2)}(c)$, density $f(c)$ of Z at c, and $\sigma_\pm^2(c) = \text{Var}(Y \mid Z = c^\pm)$. For local linear regression with bandwidth h,

$$\text{Bias} \approx C_1 h^2 \underbrace{\big(\mu_+^{(2)}(c) - \mu_-^{(2)}(c)\big)}_{\text{curvature difference}}, \qquad \text{Variance} \approx \frac{C_2}{Nh} \cdot \underbrace{\frac{\sigma_+^2(c) + \sigma_-^2(c)}{f(c)}}_{\text{noise / density at } c}.$$

As h grows the variance falls like $1/h$ but the bias grows like h^2; as h shrinks the bias falls but the variance explodes. Minimizing MSE = Bias^2 + Variance (Imbens and Kalyanaraman 2012):

$$h_{\text{MSE}} = C_K \left(\frac{\sigma_+^2(c)/f_+(c) + \sigma_-^2(c)/f_-(c)}{\big(\mu_+^{(2)}(c) - \mu_-^{(2)}(c)\big)^2}\right)^{1/5} N^{-1/5}.$$

(Setting the derivative of $C_1^2 h^4 D^2 + C_2 V/(Nh)$ to zero gives $h^5 \propto V/(ND^2)$.) The optimal bandwidth shrinks with N, but slowly. All quantities (curvatures, variances, density) are unknown and estimated, curvature by local quadratic regression.

4.4 Why the MSE-optimal bandwidth may fail for inference

h_{MSE} minimizes point-estimation error only. Keeping variance low, it picks a relatively wide window ($\propto N^{-1/5}$), so the bias is not negligible relative to the standard error: a nominal 95% CI may cover only about 80% of the time (undercoverage). Remedies: *robust bias correction* (RBC; Calonico, Cattaneo and Titiunik 2014) re-centers the interval by subtracting an estimate of the leading bias term (and accounts for its variability); the *CE-optimal bandwidth* (Calonico, Cattaneo and Farrell 2020) minimizes coverage error, the gap between nominal and actual coverage, with $h_{\text{RBC}} \propto N^{-1/4}$ for local linear fits. Rule of thumb $\hat{h}_{\text{RBC}} \approx (0.7\text{–}0.8) \times \hat{h}_{\text{MSE}}$: valid p-values need a tighter window than the point estimate.

Bandwidth	Rate	Bias	Variance	CI length	Coverage
MSE-optimal	$N^{-1/5}$	high	low	short	fails
Undersmoothing	$N^{-1/3}$	very low	high	too long	valid
CE-optimal	$N^{-1/4}$	moderate	moderate	efficient	optimal

CE-optimal with RBC is the default in modern software (`rdrobust`).

Bandwidth sensitivity. A credible RDD shows stable estimates across a range of bandwidths. In a simulation with true $\tau = 3$, small h gives low bias and high variance (wide CIs, few observations in the window), large h gives high bias and low variance.

4.5 Covariates and software

Covariates are not needed for identification (continuity at the cutoff is the identifying restriction, unlike back-door adjustment), but they can improve precision:

$$Y_i = \alpha + \beta(Z_i - c) + \tau X_i + \gamma(Z_i - c)X_i + \delta^\top V_i + \varepsilon_i,$$

with V_i pre-treatment covariates (Calonico et al. 2019). Covariates also serve as diagnostics: a jump in V_i at c is evidence against validity.

`rdrobust` (Calonico, Cattaneo, Titiunik; R/Stata) implements MSE- and CE-optimal bandwidths (default CE-optimal with RBC), robust bias-corrected CIs, covariate adjustment, clustering and fuzzy RD. In R: `library(rdrobust); rd <- rdrobust(Y, Z, c = 1500); summary(rd)`.

5 Fuzzy RD

Purpose. Treat imperfect compliance with the cutoff as a local instrumental-variable problem.

Concept. In the fuzzy design crossing the cutoff only changes the probability of treatment:

$$\lim_{z \downarrow c} P[X_i = 1 \mid Z_i = z] \neq \lim_{z \uparrow c} P[X_i = 1 \mid Z_i = z],$$

but the jump is not from 0 to 1. Reasons: exemptions and discretionary enforcement, imperfect compliance, administrative errors. Examples: a test-score cutoff determines summer-school eligibility, but not all eligible students attend; age-based eligibility with early or late applicants. Many real RD settings are fuzzy.

Fuzzy RD as IV: the Wald estimator. Take $W_i = \mathbb{1}\{Z_i \geq c\}$ as instrument for the treatment received X_i; near c, W_i is as good as randomly assigned. Under continuity,

$$\tau_{\text{FRD}} = \frac{\lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]}{\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z]} = \frac{\text{ITT effect on } Y}{\text{first-stage effect on } X},$$

the Wald estimator, i.e. IV/LATE applied locally at the cutoff.

Assumptions. (1) Continuity of $\mathbb{E}[Y_i(x) \mid Z_i = z]$ and $\mathbb{E}[X_i(w) \mid Z_i = z]$ at c; (2) first stage: $\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z] \neq 0$; (3) monotonicity: crossing c makes no one less likely to be treated (no defiers). These mirror relevance, exclusion and monotonicity; exclusion is built in, since W_i affects Y_i only through X_i for compliers near c.

Interpretation. $\tau_{\text{FRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}, Z_i = c]$, a LATE with *double locality*: it applies only to compliers (units whose treatment is actually changed by crossing the cutoff) and only at the cutoff. External validity is limited, internal validity strong; many policy questions naturally produce fuzzy RD settings.

6 Case study: remedial education in Chicago

Purpose. See a fuzzy RD in action (Jacob and Lefgren 2004, REStat).

Background. In 1996 Chicago Public Schools ended "social promotion": students had to meet minimum scores on the Iowa Test of Basic Skills (ITBS) reading and math tests to advance (e.g. third graders needed 2.8 grade equivalents in both). Students below the cutoff had to attend summer school and could be retained. *Question:* what is the causal effect of summer school and grade retention on achievement?

Design. Treatment: summer school and retention. Fuzzy, since exemptions and discretionary enforcement mean some students above the cutoff are retained and some below are not; but the probability of treatment jumps sharply at the cutoff, identifying the LATE there. Data: CPS administrative records 1993–1999, about 350,000 third and sixth graders; outcomes: changes in ITBS reading and math (in logits) one and two years later.

IV estimation. With the instrument $W_i = \mathbb{1}\{\text{Score}_i \geq \text{cutoff}\}$ (above/below the cutoff; here falling below triggers treatment, so the first-stage jump is negative),

$$\begin{aligned} \text{First stage:} \quad & \text{Treat}_i = \alpha_0 + \alpha_1 W_i + \alpha_2 \text{Score}_i + \alpha_3 (W_i \times \text{Score}_i) + \nu_i, \\ \text{Second stage:} \quad & Y_i = \beta_0 + \tau \widehat{\text{Treat}}_i + \beta_1 \text{Score}_i + \beta_2 (W_i \times \text{Score}_i) + \varepsilon_i, \end{aligned}$$

where τ is the LATE near the cutoff. Relevance: the cutoff strongly predicts treatment; exogeneity: near the threshold, being just above or below is effectively random. (Modern RD software uses local linear regression with $W_i \times \text{Score}_i$ interactions to allow different slopes on each side.)

Results. Net effect (summer school + retention): for third graders roughly 0.11–0.13 logits in the first year (about 20% of a year's typical learning), persisting after two years with some fade-out; for sixth graders no meaningful reading effect and a small, insignificant positive math effect. Retention alone: short-term reading and math gains in grade 3; negative reading and negligible math effects in grade 6. Summer school alone: positive for third graders, especially low performers. Visually, the probability of summer school/retention drops sharply at the promotion cutoff while next-year achievement also drops there, indicating a net positive effect for third graders.

Strengths and limitations. Strengths: a natural experiment via a policy cutoff; fuzzy RD with IV enables causal inference; heterogeneity by grade, gender and SES. Limitations: LATE only (students near the cutoff, compliers); no evidence on long-term (3+ year) outcomes; manual bandwidth choice without MSE- or CE-optimal selection. The study illustrates both the power of fuzzy RD and the importance of modern bandwidth methods.

7 Summary

- RDD exploits a known cutoff rule for treatment assignment.
- Identification: continuity of the potential-outcome CEFs at the cutoff, $\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]$.
- Diagnostics: outcome plot, covariate balance, McCrary density test, placebo tests.
- Estimation: local linear regression with kernel weights.
- Bandwidth: MSE-optimal ($N^{-1/5}$) for point estimation, CE-optimal ($N^{-1/4}$) for inference; rule of thumb, shrink h_{MSE} by 20–30%.
- Fuzzy RD = IV at the cutoff, identifying the LATE for compliers at c.

- Case study: positive effects of remedial education for Chicago third graders, little effect for sixth graders.

A Kink RD

Sharp kink RD looks for discontinuities in the first derivative (slope) of $\mathbb{E}[Y_i \mid Z_i = z]$ at the threshold rather than in its level (Card, Lee, Pei and Weber 2015). Estimation is similar, but local quadratic regression is preferable.

B Assumptions for the MSE-optimal bandwidth

(Imbens and Kalyanaraman 2012.) (1) $\{Y_i, Z_i\}$ i.i.d.; (2) the density $f(z)$ is continuous with $f(c) > 0$; (3) $\mu(z)$ is three times differentiable near c; (4) K is symmetric, bounded, with compact support; (5) $\sigma^2(z)$ is continuous near c with $\sigma^2(c) > 0$.

C Regularization of the bandwidth

The optimal bandwidth has the squared curvature difference in its denominator; if the estimated curvature is near zero, h explodes. IK (2012) add a penalty $\hat{r}$ (the variance of the curvature estimate) to the denominator:

$$\hat{h}_{\text{reg}} \propto \left(\frac{\text{sum of variances}}{f(c)\,[(\text{curvature})^2 + \text{regularization}]} \right)^{1/5},$$

ensuring a finite bandwidth even for flat data. Implementation: estimate $\hat{\mu}_{\pm}^{(2)}(c)$ by local quadratic regressions and set

$$\hat{r}_+ = \frac{2160\,\hat{\sigma}_+^2(c)}{N_{h,+}h^4}, \qquad \hat{r}_- = \frac{2160\,\hat{\sigma}_-^2(c)}{N_{h,-}h^4},$$

so the penalty grows with less data (N_h) or more noise (σ^2).

D Why local linear? Polynomial order

RD uses local polynomial regression of order p. Higher p reduces bias but increases variance (more parameters from limited data near c); higher p has faster theoretical rates, but the variance cost dominates in finite samples. Division of labour: local linear ($p = 1$) for the main estimate (good boundary bias–variance balance); local quadratic ($p = 2$) for bias estimation in the MSE-optimal bandwidth (IK 2012) and in robust bias correction (CCT 2014). "Estimate with p, correct bias with $p + 1$" is the `rdrobust` default.

E Global polynomials: a cautionary tale

An older practice fitted global high-order polynomials (cubic, quartic, ...) over the whole range of the running variable. Gelman and Imbens (2019) show this is problematic: estimates are sensitive to observations far from the cutoff, high-order polynomials oscillate wildly (Runge-type behavior), and confidence intervals have poor coverage. Local polynomials fitted within h of c do not suffer from these problems. Modern best practice is local linear estimation with a data-driven bandwidth. (Further reading: *The Effect*; *The Mixtape*.)

F Source coverage

The source deck has 77 physical pages (62 numbered slides).

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